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Internship Report

Flow Analysis and Modal Decomposition of a NACA0015 Airfoil Under Reverse Flow Configuration Using PIV Data

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Executive Summary

This internship was performed at the Flow Control Laboratory under the Experimental Aerodynamics Group at the Indian Institute of Technology Kanpur, under the supervision of Asst. Prof. Dr. Tufan K. Guha. The internship involved working with the already acquired Particle Image Velocimetry (PIV) data. The data was part of an ongoing research effort on the effects of sinusoidal trailing edges on reverse flow control in high speed rotorcraft applications by Ranga Srinivas Gokul and Asst. Prof. Dr. Tufan Kumar Guha. The experiments were performed prior to the internship, and experimental setup and data acquisition were not the part of the internship.

The internship was focused on postprocessing and analysis of the experimental PIV data, with emphasis on uncovering the flow physics and flow structures from the obtained velocity fields. The work involved processing raw PIV image pairs, analyzing the resulting flow structures, and reconstructing velocity fields using flow diagnostics tools and modal decomposition techniques. Efforts were directed toward identifying regions of flow separation, vortical structures, and dominant coherent patterns in the reverse flow regime.

MATLAB based tool PIVlab and Python were used extensively for data processing and analysis. Mean flow fields were computed, and quantities such as vorticity, Q-criterion, and Reynolds stresses were evaluated and visualized from the velocity field. Proper Orthogonal Decomposition (POD) was implemented to identify dominant modes and the flow field was reconstructed using a reduced number of modes. Dynamic Mode Decomposition (DMD) was studied at a theoretical level but was not applied to the dataset due to insufficient temporal resolution of the PIV experimental data. The available data were sampled at 4 Hz, whereas literature indicates that dominant unsteady phenomena in airfoil in reverse flow configurations occur at frequencies up to the order of 1 kilohertz.

Presence of flow separation and vortical structures was observed over the baseline symmetric airfoil NACA0015, operating in reverse flow at 12° angle of attack and 1.35×10^5 Reynolds number. The modal decomposition helped to identify the dominant coherent structures present in the flow. The results illustrates the significance of PIV experiments, analysis of their data, and modal decomposition techniques. The dataset utilized here can be used to better understand the unsteady aerodynamic phenomena in reverse flow regimes. The obtained results provides insights that may be useful to develop better flow control techniques.

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List of Symbols

Symbol	Description
U, V	Mean velocity components in x, y directions
u', v'	Velocity fluctuations
ω_z	Vorticity
Q	Q criterion (Second invariant of velocity gradient tensor)
λ_2	Vortex identification criterion
c	Airfoil chord length
U_∞	Free-stream velocity
Re_c	Reynolds number based on chord
$a_i(t)$	POD temporal coefficients
ϕ_i	POD spatial modes

Chapter 1

Introduction

For a significant period in the development of experimental fluid dynamics, it was primarily focused on the study of steady and time-averaged flow features. While steady flow analyses are crucial to study mean flow features, they are unable to capture the inherently unsteady nature of many practical flows. Advances in experimental techniques, high speed photography, high frequency sensors, and computational resources have made it possible to perform detailed investigation of unsteady aerodynamic phenomena. This has revealed flow physics that used to remain hidden in time-averaged flow representations.

Unsteady aerodynamics plays an important role in determining aerodynamic performance, stability, and noise generation in a wide range of applications, such as aircraft wings, rotorcraft blades, and wind turbines. The formation, interaction, and development of coherent structures like vortices and shear layers govern the flow separation, wake dynamics, and unsteady loading. Unsteady effects can significantly influence aerodynamic behavior of airfoils operating at low to moderate Reynolds numbers, so studying them is very important. In the context of rotorcraft aerodynamics, portion of rotor blade experience reverse flow at higher forward flight speed. Under such conditions, the geometric trailing edge of the blade effectively becomes aerodynamic leading edge. This leads to early flow separation resulting in increased drag, negative lift generation, and high pitching moment impulses, which adversely affect the aerodynamic efficiency and the structural health of the rotor blades which have been well established in rotorcraft literature. The PIV dataset analyzed in this work originates from a broader experimental effort conducted by Experimental Aerodynamics Group at Flow Control Laboratory, IIT Kanpur to mitigate the adverse effects of reverse flow by assessing passive flow control strategies using serrated trailing edge.

Particle Image Velocimetry (PIV) is a non-intrusive optical measurement technique used to obtain instantaneous velocity measurements. Due to its ability to provide high resolution instantaneous flow field data over the whole 2D or 3D flow domain, PIV has become a widely used diagnostic tool in experimental aerodynamics. In the present study, already produced PIV datasets from wind tunnel experiments conducted at IIT Kanpur were analyzed. The experimental setup and data acquisition were not part of the internship scope. Instead, the focus of this work was placed on post-processing, analysis, and physical interpretation of the measured velocity fields which are very important and transferrable set of skills.

Although instantaneous PIV measurements provide detailed snapshots of the flow field, identifying dominant flow features and coherent structures directly from raw velocity data can be challenging task. Modal decomposition techniques offer a framework to dissect the flow field

and extract organized flow patterns from complex datasets. Proper Orthogonal Decomposition (POD) decomposes the flow field into a set of orthogonal spatial modes ranked by their energy content. This facilitates the identification of dominant coherent structures. While POD is effective in capturing energetically significant features, it isn't concerned with the temporal frequencies or growth and decay rates of flow structures. These information can be obtained using Dynamic Mode Decomposition (DMD) when sufficiently time resolved data are available.

The primary objectives of this work are as follows:

- To post process raw PIV image pairs and extract instantaneous velocity fields.
- To analyze derived flow quantities like mean velocity, vorticity, Reynolds stress and vortex identification metrics and identify flow separation, wake behavior, and coherent structures.
- To perform Proper Orthogonal Decomposition on the velocity field data, examine the dominant modes, and reconstruct the flow field using a fewer modes.

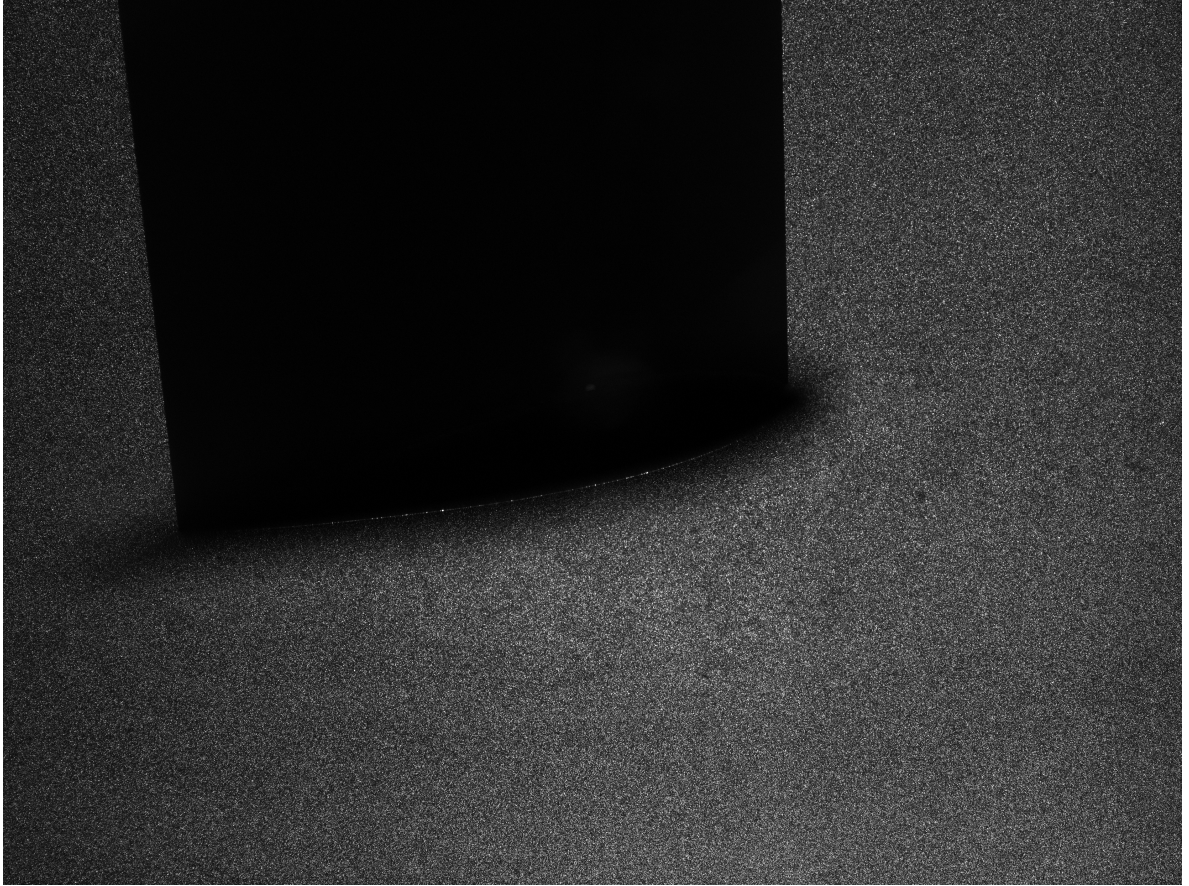


Figure 1.1: NACA 0015 airfoil geometry and PIV measurement plane.

Chapter 2

Background and Theory

Particle Image Velocimetry is a non-intrusive optical method for flow visualization that measures fluid flow velocities by analyzing the motion of seeded particles within the flow. It provides instantaneous, spatially resolved velocity fields that form the basis for the fluid flow analysis. The fluid is seeded with small tracer particles selected on the basis of stokes number such that they faithfully follow the flow dynamics. A light sheet is created using laser to illuminate the particles in the desired plane of flow. Pair of images are taken at small time interval using high speed camera. High speed camera is often synchronized with pulsed laser using a synchronizer for high speed flows to avoid blurry image and act as photographic flash. The time delay between the laser pulses should be long enough to capture the displacement of the tracer particles and short enough so that particles with an out-of-plane velocity component don't leave the light sheet. The motion of particles in pair of frame are analyzed by PIV software using cross correlation algorithm to obtain the instantaneous displacement field and velocity field using time interval between the pair images. While velocity field contains detailed information about the flow, evaluation of derived quantities can be necessary to highlight important flow features and gain meaningful physical interpretation. Quantities such as mean velocity, vorticity, vortex identification metrics, and Reynolds stresses are often used to characterize flow structures, separation, and unsteady behavior. These fields were evaluated from measured velocity field using algebraic relations and spatial gradients in this work.

Basic Flow Quantities

$$\begin{aligned}u_{\text{mean}} &= \frac{1}{n} \sum_{t=1}^n u_t, \\v_{\text{mean}} &= \frac{1}{n} \sum_{t=1}^n v_t, \\U_{\text{mean}} &= \sqrt{u_{\text{mean}}^2 + v_{\text{mean}}^2}, \\\omega &= \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}.\end{aligned}$$

The mean velocity field gives a time averaged representation of the flow field and is crucial to identify global flow features like separation regions, reattachment bubbles, wakes, etc. Vorticity highlights the local flow rotation and helps to identify shear layers and vortical structures. Though for the vortex identification, vorticity alone is not sufficient.

2.1 Vortex Identification Methods

Vorticity should be supplemented by vortex identification metrics to distinguish rotational motion from strain-dominated regions in the flow. This can be done by employing robust vortex identification metrics that isolates vortical structures from the flow. The velocity gradient can be decomposed into a symmetric strain rate tensor S and an antisymmetric rotation tensor Ω . This decomposition allows to extract the information about region where rotation dominates the strain rate.

Vortex Identification Criteria

$$Q = \frac{1}{2}(\|\Omega\|^2 - \|S\|^2),$$

where Q is positive.

$$\lambda_2 = \text{second eigenvalue of } (S^2 + \Omega^2),$$

where λ_2 is negative.

The Q criterion identifies vortical regions as regions in flow where the local rotation dominates over the strain. It is based on second invariant of velocity gradient tensor. The λ_2 criterion identifies vortices as regions where the pressure is locally minimized in plane normal to vortex axis. It is also Galilean invariant like Q . λ_2 is more geometric, relying on eigenvalue analysis, whereas Q Criterion is more physical, emphasizing rotational dominance. Other vortex identification metrics exists as well, but these two are by the far most widely used.

2.2 Proper Orthogonal Decomposition

Proper orthogonal decomposition (POD) is a data driven modal analysis technique that enables a reduction in the complexity of high dimensional datasets while retaining key features and significant patterns. In fluid dynamics, POD is widely used to extract coherent structures from velocity, pressure, or vorticity fields obtained from experimental or numerical data. It decomposes a flow field into a set of orthogonal spatial modes and corresponding temporal coefficients. For the given number of modes, it captures the maximum possible kinetic energy of the flow. Because of this, dominant large scale flow features are represented by lower order modes which have higher energy content, while the higher order modes represent small scale fluctuations and have less energy content. Mathematically, It is closely related with Principal Component Analysis (PCA) in statistics and Singular Value Decomposition (SVD) in linear algebra. When it is applied to a matrix constructed from flow field data, the matrix is decomposed into singular values, left singular vectors, and right singular vectors. The left singular vectors represent spatial modes often called eigenflow fields or eigen modes while the right singular vectors represent temporal evolution is referred as eigen time series. The singular values quantify the energy content of each mode ordering them in descending order based on their contribution to total flow energy.

Lower order modes represent dominant large scale flow structure whereas the higher order modes capture smaller scale less energetic fluctuations. It is well suited for data analysis of experimental data as it doesn't require time resolved sampling and is less susceptible to experimental noise. In this work, POD is employed to identify dominant flow structures from the experimentally obtained PIV dataset and to reconstruct the flow field with fewer number of modes. This enables efficient representation and interpretation of flow field data.

2.3 Dynamic Mode Decomposition

Dynamic Mode Decomposition (DMD) is a modal decomposition technique that decomposes flow field data into modes with distinct frequencies and corresponding growth or decay rates. DMD algorithm combines the characteristics of proper orthogonal decomposition (POD) and Fourier transform. Unlike POD, DMD does not require modes to be orthogonal and directly associates spatial structure with temporal dynamics. DMD requires flow field data to be sufficiently time resolved. This ensures that the identified modal frequencies and growth rates are accurate and capture dominant unsteady features. In this work, DMD is discussed and studied only at a conceptual level to draw contrast with POD. It is not implemented here as the available PIV data were sampled at a frequency insufficient to resolve the unsteady phenomena reported in the literature for symmetric airfoil reverse flow configurations.

Chapter 3

Methodology

This work focuses on the post processing, analysis, and modal decomposition of experimentally obtained PIV data. Raw image pairs were processed using PIVlab, a Matlab based free and open source GUI Particle Image Velocimetry software, to extract the instantaneous velocity field data. The extracted velocity field data were subsequently imported in python to obtain derived and mean field. Proper Orthogonal Decomposition was performed on the unsteady component of the velocity field data.

3.1 Overview of Dataset

The dataset analyzed in this work was obtained from an ongoing research project on the effect of Sinusoidal Trailing Edge for the Control of Reverse Flow on High-Speed Rotorcraft conducted by the Experimental Aerodynamics Group (EAG), Flow Control Laboratory, IIT Kanpur. The group had performed experiments on subsonic open-return and closed-return wind tunnel facility of IIT Kanpur for the measurement of aerodynamic forces and moments, tuft flow visualization, and PIV. The data for baseline configuration NACA 0015 operating in reverse flow at 12° angle of attack, 20 m/s freestream velocity, 0.1 m chord, and 1.35×10^5 Reynolds number was provided by EAG for the internship. Pairwise images were taken at $17\mu\text{s}$ time interval. The PIV data had the sampling frequency of 4 Hz. Total 500 pair of snapshots were captured. EAG, IIT Kanpur is the source and owner of this dataset and it was only analyzed here.

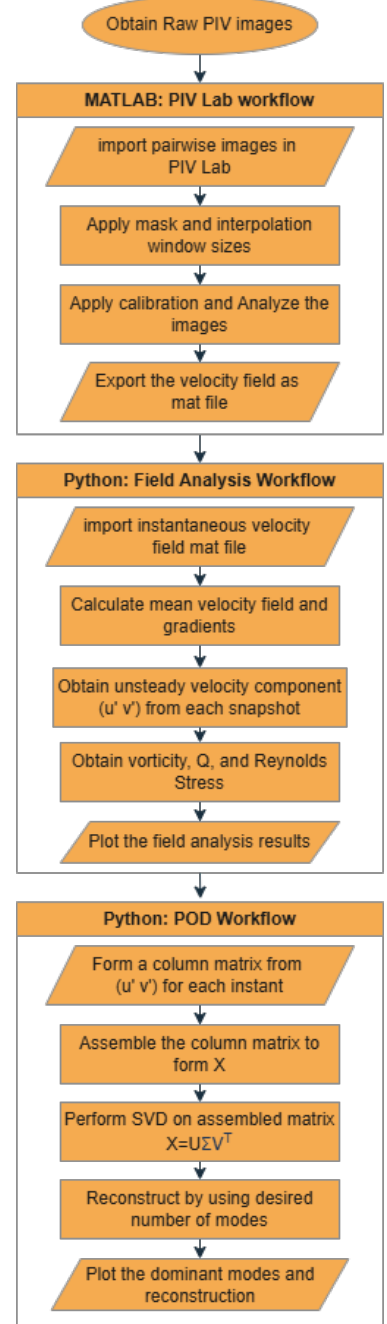


Figure 3.1: Post-processing and analysis workflow adopted using PIVlab and Python.

3.2 PIV Processing Using PIVlab

PIVlab, a Matlab based free and open source application, was used for processing raw PIV images. The images were imported as paired frames and mask was applied in the regions without PIV seeds. CLAHE was applied to enhance the contrast. A multi-pass FFT window deformation interrogation scheme was used, consisting of four interrogation area passes, two with interrogation sizes of 64 px and remaining two of 32 px. Spatial calibration was performed using a calibration image to convert pixel displacement to physical units, and time step of 0.017 ms was specified between the image pairs. This introduced conversion factor of $1 \text{ px/frame} = 3.34 \text{ m/s}$ for the image pairs. The processed dataset yielded instantaneous two-dimensional velocity fields, which were exported in `.mat` format for subsequent analysis.

3.3 Python-Based Post-Processing

Python based post processing was performed to compute mean flow quantities and derived flow metrics from instantaneous velocity fields. The mat file from the PIVlab was imported in python using `loadmat` function of `scipy` library. Additionally, `numpy` library was used extensively to perform matrix and vector operations on the imported data and `matplotlib` was used to plot the results. Firstly, the mat file was imported and `u`, `v`, `x`, and `y` were extracted from it. It was observed that each of these matrix had the shape of (500, 154, 206) representing number of time steps and number of rows and columns in each snapshot respectively. Mean method of `numpy` was used to calculate the components of the mean velocity field (`Umean`, `Vmean`) from the 500 time steps and magnitude of the mean velocity field was computed. The position matrix `x` and `y` are meshgrid so `x` varies in between the columns and `y` varies in between the rows meaning all the rows in `x` are identical and all the columns in `y` are identical. So, a row from `x` and a column from `y` are taken and difference vector is computed for both of them. Mean of the difference vector is calculated to obtain `dx` and `dy`. Gradient of `Umean` and `Vmean` is computed along both `x` and `y` axis with `dx` and `dy`. Vorticity is calculated by subtracting gradient of `Umean` along `y` from gradient of `Vmean` along `x`. The resulting vorticity is normalized by multiplying it with the ratio of chord length to freestream velocity. Similarly, `Q` is calculated using the gradients of `Umean` and `Vmean`. The unsteady velocity component calculated by subtracting the mean velocity components from the instantaneous velocity components is used to calculate Reynolds stress.

3.4 Modal Decomposition Procedure

The `numpy` library is used for performing singular value decomposition and matrix and vector operations whereas the `matplotlib` library is used for plotting results of analysis and reconstruction. Unsteady component of velocity is calculated by subtracting mean component of velocity from velocity component of each frame. Thus, 500 snapshots of unsteady velocity components

u_prime (500, 154, 206) and v_prime (500, 154, 206) are obtained. Each snapshot of unsteady velocity component u_prime(t) (154, 206) and v_prime(t) (154, 206) matrix is flattened into column vectors u_prime_col (500, 31724) and (500, 31724) containing 500 snapshot of flatten components. These column vectors are stacked vertically to obtain a matrix with 500 columns of time steps (500, 63448) called $\mathbf{X}$. All the NaNs, negative, and positive infinity are replaced with zero. This replacement was performed to ensure numerical stability and is conceptually safe and sound. An economy SVD was performed on the snapshot matrix $\mathbf{X}$ such that

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T,$$

where $\mathbf{U}$ contains the spatial POD modes, $\mathbf{\Sigma}$ contains the singular values, and $\mathbf{V}$ represents the temporal evolution of the modes. The decomposed matrices are assigned to $\mathbf{U}$ and $\mathbf{Vt}$ and mode vector to $\mathbf{S}$. A vector called energy which represent the energy contribution of each mode is calculated from mode vector normalizing it by dividing square of each mode with sum of squares of all modes. Diagonalized mode vector is premultiplied to eigen timeseries $\mathbf{Vt}$ to obtain temporal coefficients matrix $\mathbf{a}$. The matrix $\mathbf{a}$ containing number of rows equal to desired mode number can be premultiplied by spatial matrix $\mathbf{U}$ with number of columns equal to desired mode number and reshaped into the shape of original matrix $\mathbf{u}$ and $\mathbf{v}$ thus reducing the complexity of dataset.

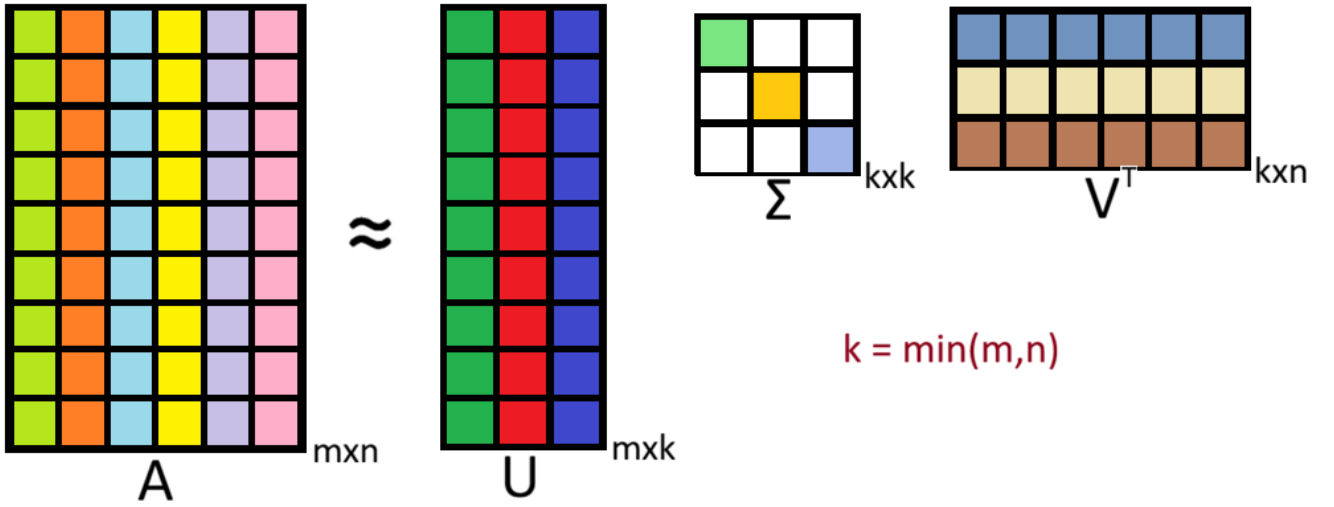


Figure 3.2: Graphical representation of matrix operation of SVD.

Chapter 4

Results and Analysis

The obtained results are analyzed and interpreted here.

4.1 Mean Velocity Field

The mean velocity field shows that the flow is completely separated at 12° angle of attack under reverse flow conditions. Flow separation occurs at the sharp leading edge and does not reattach over the chord length. The streamlines show that there is a massive recirculation region in the separated flow. The white dashed line marks the interface separating regions of negative streamwise velocity ($u < 0$) from positive streamwise velocity ($u > 0$). Along this line, the mean streamwise velocity is zero. It can also be observed that the flow just below the recirculation wake accelerates downstream from aerodynamic leading edge. Near the downstream end of the wake, the streamlines indicate the upward motion of fluid particles followed by downward deflection due to interaction with the upper shear layer.

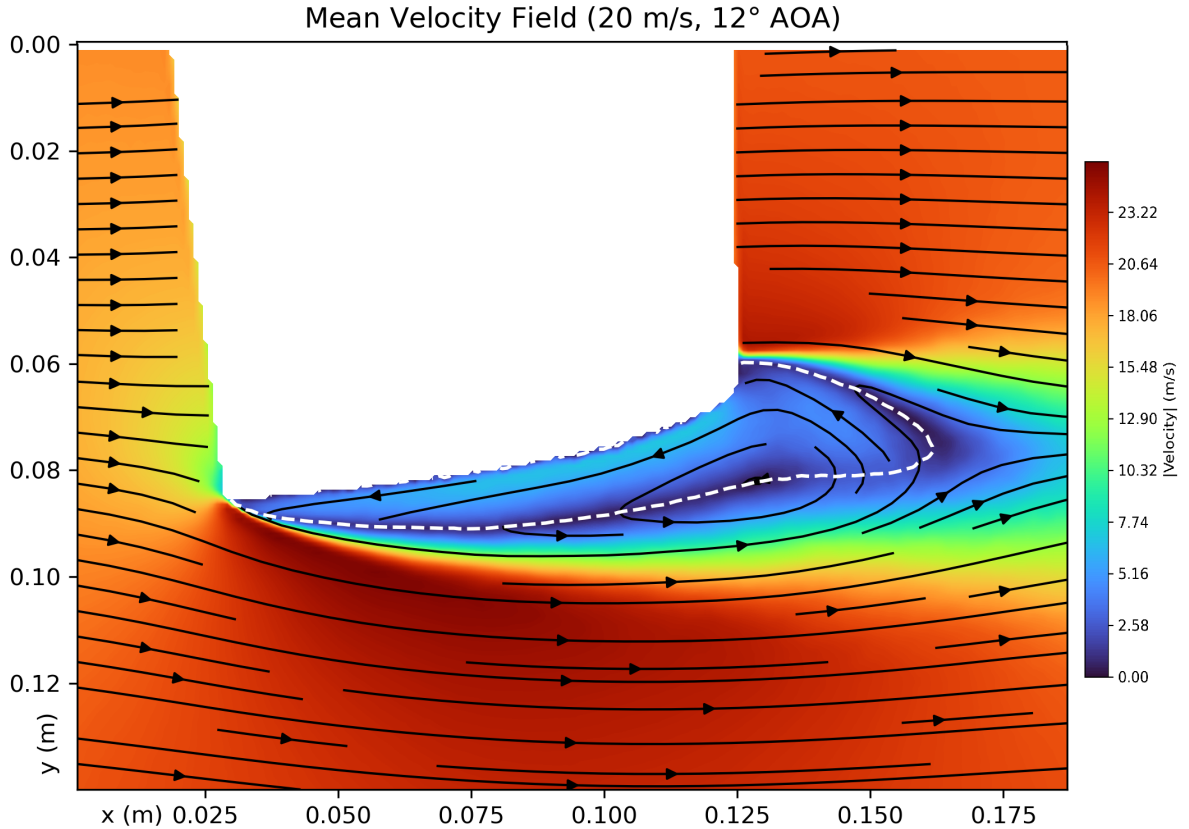


Figure 4.1: Mean velocity magnitude contour, $U_\infty = 20\text{m/s}$, $\alpha = 12^\circ$

4.2 Vorticity and Coherent Structures

The normalized vorticity field highlights the region with high rotation and shear. High vorticity contours closely follow the separated shear layers as confirmed by the streamlines. The shear layer originating from the lower surface is rotating in counter clockwise direction, while the flow exiting the upper surface near trailing edge exhibits rotation in clockwise direction. The normalized vorticity masked by high Q criterion values isolates regions where rotation dominates strain in the mean flow. It reveals the coherent vortical structures present in the flow. Strong vorticity is observed near the leading edge where flow separates from lower surface and near trailing edge where flow leaves upper surface. The abrupt flow separation produces intense velocity gradients near leading edge and the shear layers interact and shed into wake in near trailing edge.

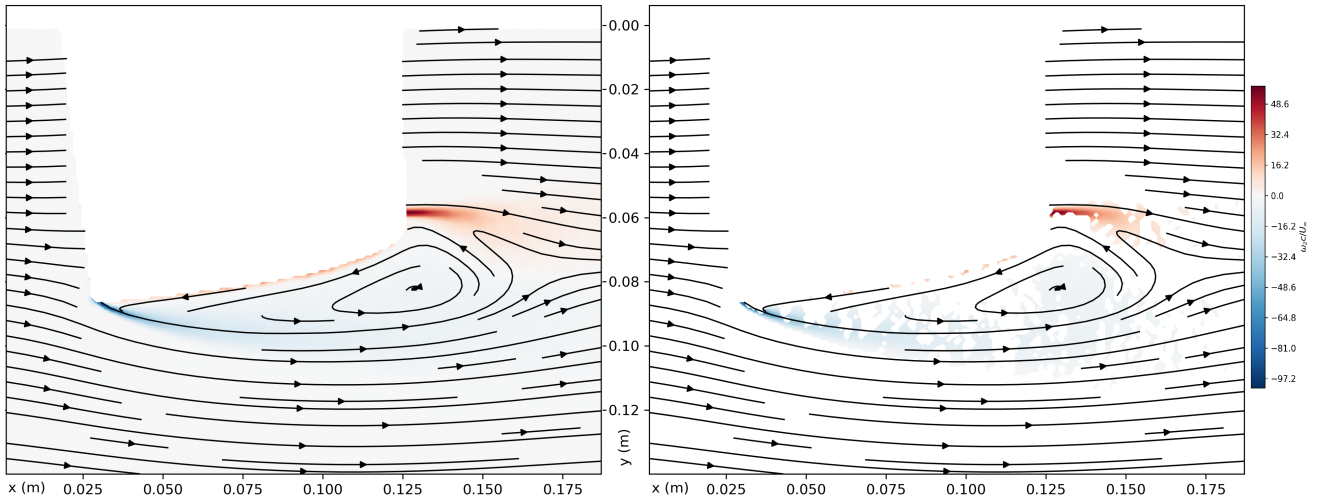


Figure 4.2: Mean Normalized Vorticity Contour without (left) and with (right) High Q mask, $U_\infty = 20m/s$, $\alpha = 12^\circ$

4.3 Turbulence Intensity

The turbulence intensity is calculated using the magnitude of the velocity fluctuations normalized by the freestream velocity. It provides a measure of the strength of fluctuating velocity field in the flow. The turbulence intensity contour shows that the wake region exhibits higher turbulence levels compared to flow in freestream. The highest turbulence intensity is found just outboard of the separation boundary ($u=0$). This behavior is expected due to the strong velocity gradients and interaction between the backward flow and forward flow region which ends up energizing the flow. In contrast, the freestream region exhibits very low turbulence intensity consistent with uniform flow condition.

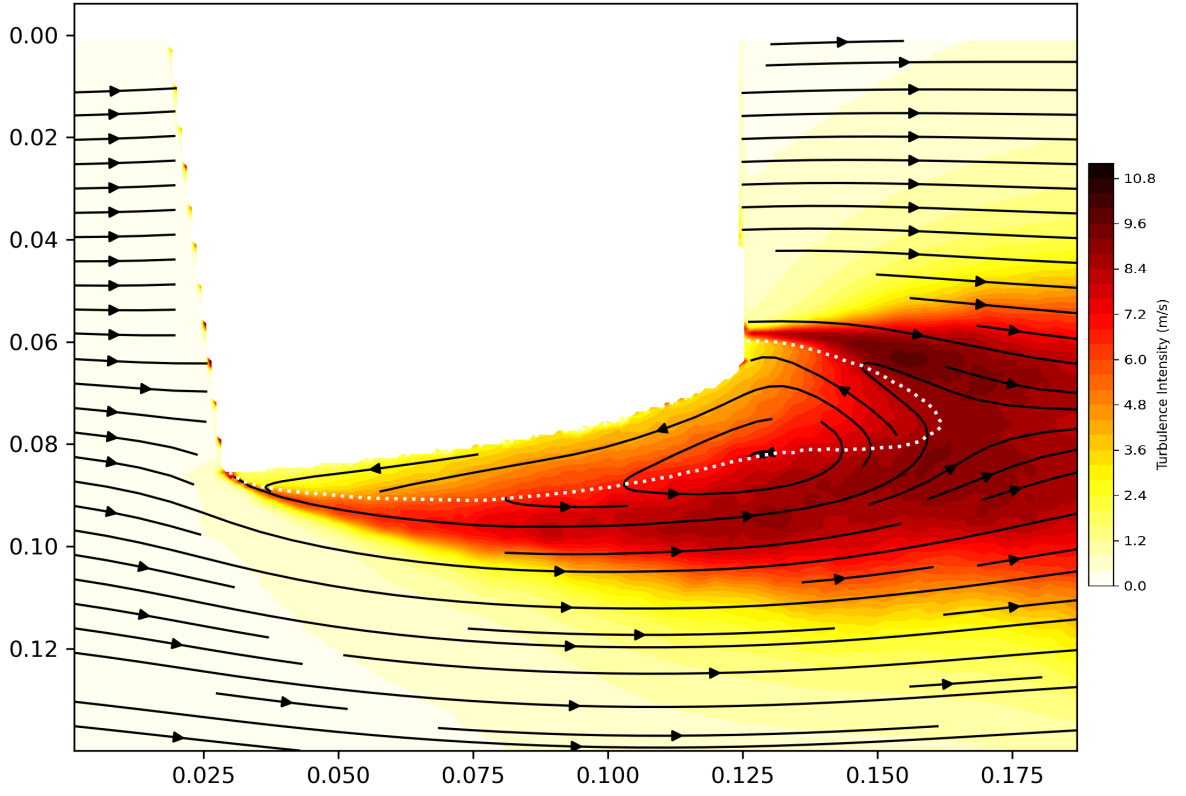


Figure 4.3: Turbulence Intensity of flow, $U_\infty = 20\text{m/s}$, $\alpha = 12^\circ$

4.4 Streamwise Normal stress and Reynolds Shear Stress

The streamwise Normal stress $\overline{u'u'}$ represents the intensity of velocity fluctuations in streamwise direction and is always positive. High values are observed in the interface of wake and freestream, where velocity gradients are strong. The Reynolds Shear Stress $\overline{u'v'}$ represents the correlation between streamwise and transverse components of velocity fluctuations and can have both positive and negative values. Regions with high shear stress are concentrated along the wake boundary. This indicates strong momentum exchange between the separated flow and nearby freestream.

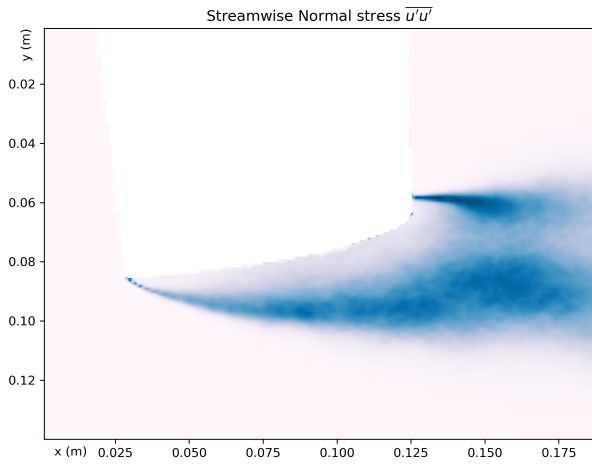


Figure 4.4: Streamwise Normal stress $\overline{u'u'}$

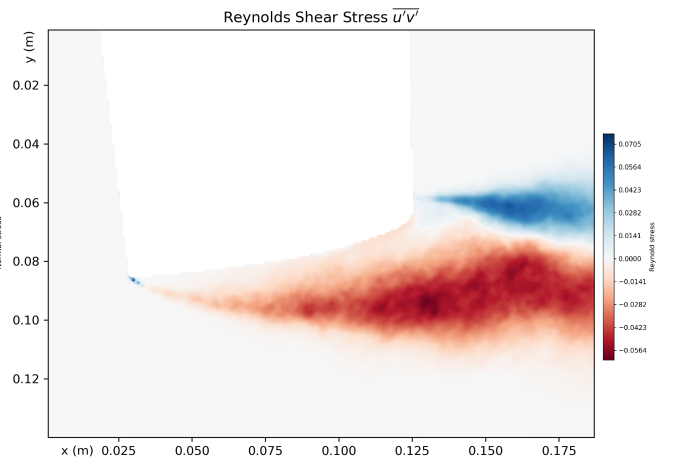


Figure 4.5: Reynolds stress $\overline{u'v'}$

4.5 POD Results

The cumulative energy versus mode number plot shows that approximately 50% of the total fluctuation energy is captured by the first 7 modes, 70% by the first 30 modes, and 90% by the first 160 modes. The curve exhibits a gradual saturation behavior illustrating slower energy decay across the modes, a characteristic of strong flow separation. The Energy fraction vs mode number plot further shows that each following mode contributes progressively less energy. Proper Orthogonal Decomposition is optimal in L^2 sense meaning no other linear decomposition method can represent the same amount of energy with fewer number of modes.

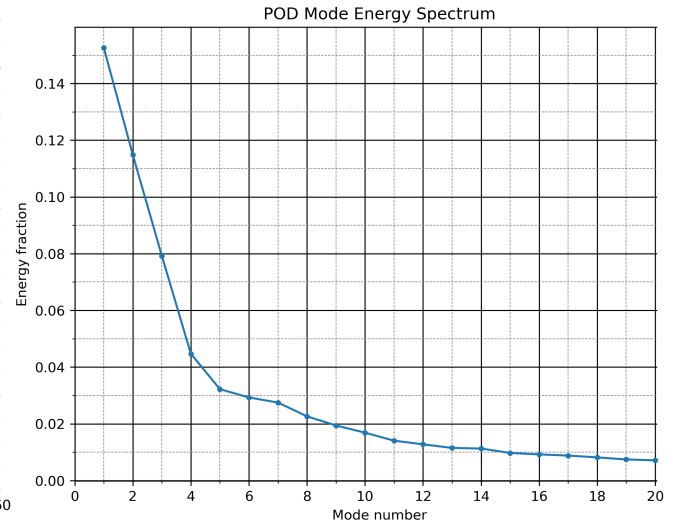
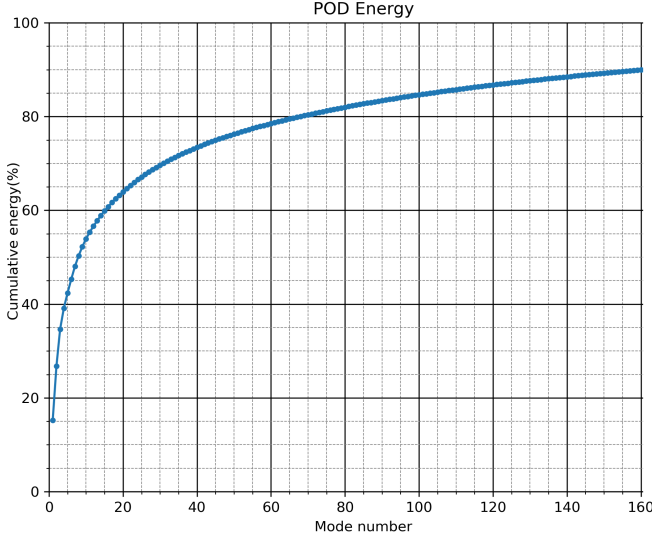


Figure 4.6: Plot of cumulative energy vs mode number
Figure 4.7: Energy fraction contribution of first 20 modes

The spatial contour of first three POD modes are shown in the following image. The first mode represents the most energetic fluctuation. The dominant structures are concentrated near the trailing edge shear layer from upper surface and in the separated wake below lower surface. The second mode exhibits similar spatial distribution with phase shifted structure. It suggests the oscillatory like spatial behavior of the separated shear layer. This can be speculated from the green region downstream the pink region in the wake below the lower surface. The third mode represent smaller scale instabilities within the separated flow and has further lower energy content.

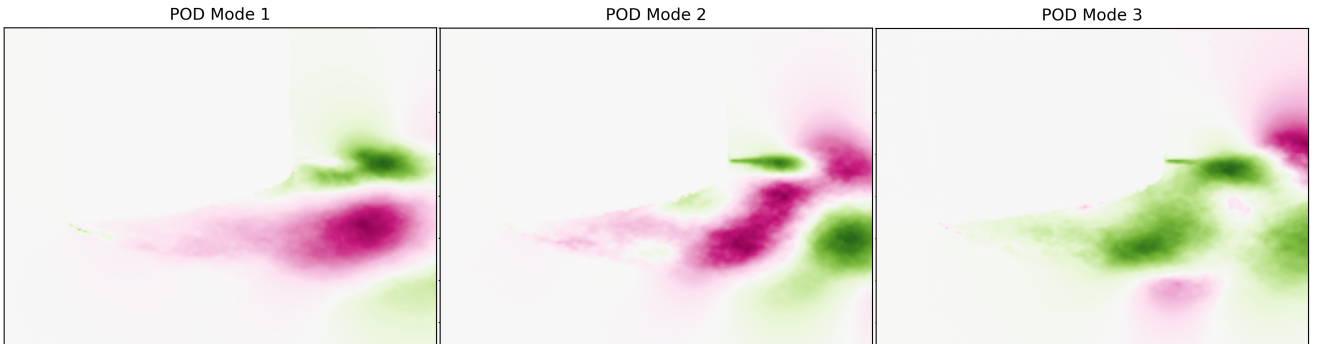


Figure 4.8: First three dominant POD spatial modes

As the flow is fully separated and has strong recirculation and chaotic wake, there are number of energy scales and the flow needs many modes to reconstruct it. It is strongly evident through following POD based reconstruction. All three images correspond to the same instantaneous frame of the velocity data. The first two images are reconstruction from POD whereas the last image is the original instantaneous field. The first image, reconstructed with first 30 modes, represents 70% of the total flow energy and appears smoother than other two images. The separation line is less jagged compared to full field reconstruction and original image. The second images uses all available modes to reconstruct the velocity field. It closely resembles original instantaneous velocity field shown in third image. This illustrates that the higher order modes are responsible for capturing finer unsteadiness present in the data.

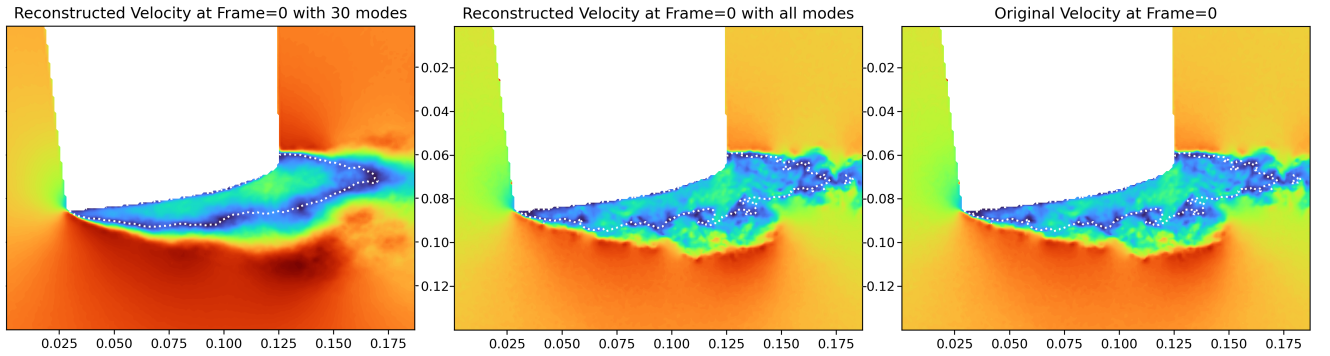


Figure 4.9: Comparison of reconstructed field with original field

Chapter 5

Conclusions

This internship opportunity provided valuable exposure to data analysis for experimental fluid dynamics within a limited duration of three and half weeks. The primary focus of the work was on postprocessing of raw PIV dataset obtained from wind tunnel experiments conducted by host laboratory and interpretation of the obtained results. Although the experimental setup and data acquisition was outside the scope of internship, the internship provided familiarity with experimental techniques such as force and moment measurements in wind tunnel and tuft flow visualization through observations and PIV through discussion. The core contribution of this work involved handling of raw PIV image pairs using PIVlab and processing it to extract instantaneous flow fields, followed by extensive postprocessing and analysis of flow field data in python. Mean flow fields, vorticity, Q criterion, Reynolds stress, and statistical measures such as velocity fluctuations were computed and analyzed to understand flow separation and wake behavior in reverse flow configuration for symmetric airfoil at high angle of attack. A significant fraction of internship was dedicated to the implementation of Proper Orthogonal Decomposition (POD) for the available dataset. POD was used to identify dominant flow structures, analyze energy distribution of modes, and reconstruct the reduced order representation of fluid flow. Dynamic Mode Decomposition (DMD) was studied at theoretical level to understand its advantages and limitations. However, it was not implemented due to insufficient temporal resolution of the available PIV measurements. Due to the short duration of internship, conducting independent experimental measurement was not feasible. Nevertheless, the internship provided with strong practical training in experimental data handling, postprocessing, modal analysis, and interpretation of unsteady flow physics. The skills and understanding gained during this internship are highly transferable in experimental fluid dynamics and across scientific domain. Detailed mathematical formulations and computational implementations are provided in the appendices.

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Appendix A

Mathematical Formulation

Mathematical relations used for the analysis and postprocessing of the Particle Image Velocimetry (PIV) data are summarized here.

Velocity Decomposition

The instantaneous velocity field obtained from PIV measurements is decomposed into mean and fluctuating components as

$$u(x, y, t) = \bar{u}(x, y) + u'(x, y, t), \quad v(x, y, t) = \bar{v}(x, y) + v'(x, y, t) \quad (\text{A.1})$$

where the bar denotes time-averaging over all snapshots.

Mean Flow and Derived Quantities

The mean velocity components are defined as

$$\bar{u}(x, y) = \frac{1}{N} \sum_{t=1}^N u(x, y, t), \quad \bar{v}(x, y) = \frac{1}{N} \sum_{t=1}^N v(x, y, t) \quad (\text{A.2})$$

The magnitude of the mean velocity is

$$\bar{U} = \sqrt{\bar{u}^2 + \bar{v}^2} \quad (\text{A.3})$$

Velocity Gradient and Q Criterion

The velocity gradient tensor is given as

$$\nabla \mathbf{u} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} \quad (\text{A.4})$$

$$\mathbf{S} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T), \quad \mathbf{\Omega} = \frac{1}{2}(\nabla \mathbf{u} - \nabla \mathbf{u}^T) \quad (\text{A.5})$$

The velocity gradient is decomposed into symmetric and antisymmetric parts. The Q criterion is defined as

$$Q = \frac{1}{2} (\|\mathbf{\Omega}\|^2 - \|\mathbf{S}\|^2) \quad (\text{A.6})$$

Reynolds Stress and Turbulence Statistics

The Reynolds stress components are defined as

$$\overline{u'u'} = \frac{1}{N} \sum_{t=1}^N u'^2, \quad \overline{v'v'} = \frac{1}{N} \sum_{t=1}^N v'^2, \quad \overline{u'v'} = \frac{1}{N} \sum_{t=1}^N u'v' \quad (\text{A.7})$$

The turbulence intensity is defined using the standard deviation of velocity fluctuations as

$$TI = \sqrt{\sigma_u^2 + \sigma_v^2} \quad (\text{A.8})$$

Proper Orthogonal Decomposition

A snapshot matrix $\mathbf{X}$ is constructed using the fluctuating velocity components:

$$\mathbf{X} = \begin{bmatrix} u'_1 & u'_2 & \cdots & u'_N \\ v'_1 & v'_2 & \cdots & v'_N \end{bmatrix} \quad (\text{A.9})$$

Each column of $\mathbf{X}$ contains a spatially flattened snapshot of the two dimensional velocity field. The snapshot matrix is decomposed using singular value decomposition as

$$\mathbf{X} = \mathbf{\Phi} \mathbf{\Sigma} \mathbf{V}^T \quad (\text{A.10})$$

The energy fraction of the k^{th} mode is computed as

$$E_k = \frac{\sigma_k^2}{\sum_{i=1}^N \sigma_i^2} \quad (\text{A.11})$$

The velocity field reconstructed using the first r modes is given by

$$\mathbf{X}_r = \sum_{k=1}^r \sigma_k \phi_k \mathbf{v}_k^T \quad (\text{A.12})$$

Appendix B

Computational Implementation

Python codes used for importing the PIV dataset, computing derived flow quantities, and performing Proper Orthogonal Decomposition (POD) are presented here. Plotting and visualization codes are not included here.

Data Loading and Preparation

```
1 import numpy as np
2 from scipy.io import loadmat
3
4 data = loadmat("./reverseflow.mat")
5
6 u = np.array([data['u_original'][i, 0] for i in range(data['u_original'].
    shape[0])])
7 v = np.array([data['v_original'][i, 0] for i in range(data['v_original'].
    shape[0])])
8 x = np.array(data['x'][0, 0])
9 y = np.array(data['y'][0, 0])
10
11 dx = np.mean(np.diff(x[0, :]))
12 dy = np.mean(np.diff(y[:, 0]))
```

Mean Flow and Gradient Computation

```
1 Umean = np.mean(u, axis=0)
2 Vmean = np.mean(v, axis=0)
3 Umag = np.sqrt(Umean**2 + Vmean**2)
4
5 dUdy, dUdx = np.gradient(Umean, dy, dx)
6 dVdy, dVdx = np.gradient(Vmean, dy, dx)
```

Vorticity and Vortex Identification

```

1 c = 0.1
2 U_inf = 20.0
3
4 omega_z = dVdx - dUdy
5 omega_norm = omega_z * (c / U_inf)
6
7 Q = 0.5*((dVdx - dUdy)**2 - (dUdx**2 + dVdy**2 + 2*(0.5*(dUdy + dVdx))**2))
8
9 Q_thresh = 2e5
10 mask = Q > Q_thresh
11 omega_masked = np.where(mask, omega_norm, np.nan)

```

Turbulence Statistics

```

1 u_prime = np.array([u[i]-Umean for i in range(u.shape[0])])
2 v_prime = np.array([v[i]-Vmean for i in range(v.shape[0])])
3
4 u_std = np.std(u, axis=0)
5 v_std = np.std(v, axis=0)
6
7 turb_intensity = np.sqrt(u_std**2 + v_std**2)
8
9 UU_unsteady_mean = np.mean(u_prime * u_prime, axis=0)
10 UV_unsteady_mean = np.mean(u_prime * v_prime, axis=0)

```

POD Implementation

```

1 nt, ny, nx = u.shape
2 npts = ny * nx
3
4 u_prime_col = u_prime.reshape(nt, -1).T
5 v_prime_col = v_prime.reshape(nt, -1).T
6
7 X = np.vstack([u_prime_col, v_prime_col])
8 X = np.nan_to_num(X)
9
10 U, S, Vt = np.linalg.svd(X, full_matrices=False)
11
12 Phi = U
13 a = np.diag(S) @ Vt
14
15 energy = S**2 / np.sum(S**2)
16 cum_energy = np.cumsum(energy)

```

Flow Field Reconstruction

```
1 def reconstruct_all(k):
2     Phik = Phi[:, :k]
3     ak = a[:, k, :]
4     Xp = Phik @ ak
5
6     u_rec = Xp[:npts, :].T.reshape(nt, ny, nx) + Umean
7     v_rec = Xp[npts:, :].T.reshape(nt, ny, nx) + Vmean
8     return u_rec, v_rec
9
10
11 def reconstruct_snapshot(k, t):
12     Phik = Phi[:, :k]
13     ak = a[:, k, t]
14     Xp = Phik @ ak
15
16     u_snap = Xp[:npts].reshape(ny, nx) + Umean
17     v_snap = Xp[npts:].reshape(ny, nx) + Vmean
18     return u_snap, v_snap
```